

LESSON 8-3 • SESSION 1 • PRACTICE • FORM A

Multiplication Equations

Learning target – I can solve a multiplication equation by dividing both sides by the same number. · 6.EE.B.7

REMEMBER

- The **coefficient** is the number next to the variable. $5b$ means $5 \times b$.
- Division undoes multiplication. **Divide both sides** by the coefficient.
- Check** by multiplying: coefficient $\times$ your answer = the total.

WORKED EXAMPLE

5 boxes each hold the same number of pencils, 45 in all. Solve $5b = 45$.

Step 1 Divide both sides by 5: $5b \div 5 = 45 \div 5$.

Step 2 Simplify: $b = 9$.

Step 3 Check: $5 \times 9 = 45$.

Answer Each box holds 9 pencils.

WORDS coefficient (*coeficiente*) · inverse operation (*operación inversa*) · Division Property of Equality (*propiedad de división de la igualdad*)

PRACTICE

show your work · label units

1. Solve each equation.

a. $12k = 60$ ____

b. $9m = 117$ ____

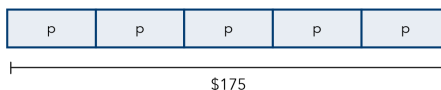
c. $11w = 88$ ____

d. $4y = 10$ ____

e. $15t = 90$ ____

f. $2r = 7$ ____

2. **Soccer Cleats.** A team spent \$175 on 5 pairs of cleats. Each pair cost the same, p dollars.



a. Write a multiplication equation. _____

b. Solve it. _____

c. Check by multiplying. _____

3. **Bookshelf Rows.** A bookshelf holds the same number of books on each of its 7 shelves. It holds 133 books in all. Write and solve an equation for b , the books on one shelf.

Hint Shelves $\times$ books on one shelf = all books.

Show your work.

Answer _____ books

4. **Egg Cartons.** A farm fills 15 cartons with e eggs each, 270 eggs in all. Solve $15e = 270$.

Hint Divide both sides by 15.

Show your work.

e = _____

5. **Training Miles.** A runner ran m miles each day for 14 days, 154 miles in all. Solve $14m = 154$.

Show your work.

m = _____

6. **Car Wash.** A car wash charges the same price for every car. Washing 24 cars brought in \$336. Write and solve an equation for the price of one wash.

Hint The number of cars is the coefficient.

Show your work.

Price _____

7. **Field Day Water.** A crew fills 18 identical coolers. Together they hold 216 cups of water. Write and solve an equation for the cups in one cooler. Check it.

Hint Check: $18 \times$ your answer should be 216.

Show your work.

Answer _____ cups

8. **Concert Rows.** A hall has r rows with 24 seats in each row. It has 648 seats in all.

- Write and solve an equation for the number of rows. _____
- Ten rows are removed for a stage. How many seats are left? _____
- How can you check part a without dividing again? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Nadia's Work.** To solve $5x = 65$, Nadia multiplied both sides by 5. She wrote $x = 325$. Explain her mistake. Then solve correctly.

TEST PRACTICE

choose the best answer

10. Solve $8n = 96$.

A $n = 88$ **B** $n = 768$ **C** $n = 12$ **D** $n = 104$

11. In $4y = 28$, the number 4 is the –

A variable **B** coefficient **C** solution **D** total

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

35 P	18 O	19 R	27 T	11 D	12 C	140 Z	\$14 U	\$312 X	875 Q
#2b		#3	#4	#5	#6	#7	#8a		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____